Akoizo Original Practice - Form 5

Division C | Mission Possible | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) An IR action has the transmitter 15 cm from the receiver. Does it qualify?

A. No; they must be at least 20 cm apart

B. Yes; any distance works

C. Yes, if the LED is visible light

D. No; they must be 50 cm apart

Answer: _____

2. (2 points) Which lever class has the load between the fulcrum and the effort?

A. Second class

B. First class

C. Third class

D. None

Answer: _____

3. (2 points) What is the first tiebreaker?

Unit: Where the points are: building a scoring plan.
Worked-case data: A Regional device starts on the first coin, completes 8 scorable actions and the Final Action at 58 s, earns all ASL points, sets up in 20 minutes, has no adjustments, and measures 60 x 60 x 70 cm. Estimate its score.
Separate investigation observations:
Plan A: 12 actions, works in 6 of 10 test runs.
Plan B: 9 actions, works in 10 of 10 test runs.
Failure cost: A failed run usually loses the Final Action, 75 no-adjust bonus, and some time points.

A. Fewest penalty points

B. Smallest device

C. Fastest setup

D. Most scorable actions

Answer: _____

4. (2 points) What must a State team do to earn water timer bonus points?

Unit: Timing: Target Time, the water timer, and pacing.
Worked-case data: Target Time 60 s. The device runs 67 s with a 25-s water timer that ended at 40 s. Calculate time points and timer bonus.
Separate investigation observations:
Timer calibration: 150 mL -> 18 s; 300 mL -> 30 s; 450 mL -> 39 s.
Other actions: Total 46 s without the timer.
Target Time: 80 s announced at State.

A. Run the timer past the Target Time

B. Use an electric pump

C. Demonstrate how the timer is adjusted for the Target Time

D. Use a scorable action as the timer

Answer: _____

5. (4 points) How many moles of CO_2 are produced by 4.2 g of baking soda with excess vinegar?

Answer: _____
_____
_____

6. (4 points) Why must rejected coins be caught inside the device?

Unit: The Start Action: coins, slots, and selective triggers.
Worked-case data: Design a rail gap that rejects nickels and dimes but passes quarters to a trigger, allowing 0.5 mm tolerance each side.
Separate investigation observations:
Gap 21.5 mm: Nickel sometimes stuck on the rail and triggered the start.
Gap 22.8 mm: Dime and nickel fell through 20 of 20; quarter carried 20 of 20.
Gap 24.0 mm: Quarter fell through in 6 of 20 trials.

Answer: _____
_____
_____

7. (4 points) Why must the gum be removed after it lands?

Unit: The Final Action, the ASL, and competition day.
Worked-case data: Estimate the landing speed of a gum pack dropped 55 cm, and suggest how to keep it on the platform.
Separate investigation observations:
Drop tests: Gum fell 52 cm; landed on platform 7 of 10 times, bouncing off 3.
Change: Added a level release shelf; dropped 51 cm.
New tests: Landed and stayed 10 of 10 times.

Answer: _____
_____
_____

8. (4 points) Explain why lower resistance nichrome cuts line faster with the same battery.

Answer: _____
_____
_____

9. (4 points) Why does a first-class lever not always give a mechanical advantage greater than 1?

Answer: _____
_____
_____

10. (6 points) Solve this case and explain your reasoning: A Regional device starts on the first coin, completes 8 scorable actions and the Final Action at 58 s, earns all ASL points, sets up in 20 minutes, has no adjustments, and measures 60 x 60 x 70 cm. Estimate its score.

Answer: _____
_____
_____

11. (6 points) Solve this case and explain your reasoning: Target Time 60 s. The device runs 67 s with a 25-s water timer that ended at 40 s. Calculate time points and timer bonus.

Answer: _____
_____
_____

12. (2 points) What explains the results?

Supplied synthetic observations:
Test 1: 1.0 g baking soda + 30 mL vinegar: balloon expanded 14 cm.
Test 2: 2.0 g + 30 mL vinegar: expanded 21 cm.
Test 3: 3.0 g + 30 mL vinegar: expanded 22 cm; undissolved powder left over.

A. Vinegar became the limiting reactant above about 2 g of baking soda

B. More baking soda always gives more gas

C. The balloon was too large

Answer: _____

13. (2 points) Define “Coin Drop Order” in the context of Mission Possible.

Answer: _____
_____
_____

14. (2 points) Define “Final Action” in the context of Mission Possible.

Answer: _____
_____
_____

15. (2 points) Define “Photodiode” in the context of Mission Possible.

Answer: _____
_____
_____

16. (2 points) Define “Sodium bicarbonate” in the context of Mission Possible.

Answer: _____
_____
_____

17. (2 points) Define “Supporting segments” in the context of Mission Possible.

Answer: _____
_____
_____

18. (2 points) Define “Action Sequence List (ASL)” in the context of Mission Possible.

Answer: _____
_____
_____